2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT)
Karnataka, India. Sep 12-13, 2025

2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT) | 979-8-3315-3200-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICERECT65215.2025.11377957

Hybrid Statistical and Deep Feature-Based Detection of Steganography in DCT and LSB Domains

D. Navya
Department of Computer Science and Engineering,
Amrita School of Computing,
Bengaluru
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie23050@bl.students.amrita.edu

Swati V
Department of Computer Science and Engineering,
Amrita School of Computing,
Bengaluru
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie23031@bl.students.amrita.edu

Penumarthi Hima Varshini
Department of Computer Science and Engineering,
Amrita School of Computing,
Bengaluru
Amrita Vishwa Vidyapeetham, India.
bl.en.u4aie23046@bl.students.amrita.edu

Neetu Srivastava
Associate Professor,
Department of Mathematics,
Amrita School of Engineering,
Bengaluru
Amrita Vishwa Vidyapeetham, India.
s_neetu@blr.amrita.edu

***Abstract*—Steganography facilitates secret communication by hiding information in digital media, and this presents great challenges to digital forensics and information security. This paper introduces a single steganalysis framework that can identify steganographic content hidden in spatial (Least Significant Bit, LSB) and frequency (Discrete Cosine Transform, DCT) domains. We derive discriminative features through statistical means and use dimensionality reduction algorithms like PCA, SVD, and Deep Matrix Factorization to improve representation. These features are then classified through both traditional and deep learning-based models. For further detection accuracy improvement, ensemble learning techniques including a strong stacking classifier are used. Statistical hypothesis testing is used to confirm cover and stego feature separability. Experimental findings show that though individual classifiers result in moderate performance, ensemble-based methods greatly enhance performance, especially for DCT-based steganography. Further, a pipeline for real-time inference is incorporated for deployment in practice. The contribution of the work is in showcasing the value of integrating machine learning with statistical domain expertise towards high-accuracy and interpretable steganalysis.**

***Index Terms*—Steganography, Steganalysis, LSB, DCT, Autoencoder, PCA, SVD, Deep Matrix Factorization, Stacking Classifier, Statistical Testing, Image Processing.**

I. Introduction

Steganography, the art of concealing confidential data inside a cover medium like images, audio, or video, is an effective tool for maintaining data privacy and security. It has several applications, especially in secure communication, where the problem is inserting data without notice. The quality of steganography techniques is based on balancing payload capacity, visual quality, and detectability. Of the various methods employed, Least Significant Bit (LSB) and Discrete Cosine Transform (DCT) are two popular methods in use. LSB provides simplicity and very high capacity but can be easily attacked, whereas DCT is more secure and provides improved perceptual quality, which can be employed in secure applications.

The paper delves into LSB and DCT-based steganography methods, emphasizing the extraction of significant features for the purpose of hidden information detection. Feature extraction techniques such as Singular Value Decomposition (SVD) and Autoencoder are employed, and the obtained features are used as inputs to different machine learning classifiers such as Logistic Regression, SVM, and ensemble techniques. Statistical tests like Chi-square, Kullback-Leibler (KL) divergence, T-tests, and McNemar's test are employed to contrast the performance of these classifiers in distinguishing stego and cover images in order to assess the efficacy and accuracy of each method under various steganographic conditions.

In multimodal steganography, methods such as Least Significant Bit (LSB) and Discrete Cosine Transform (DCT) are commonly used. LSB offers simplicity and good capacity but has low robustness and is subject to steganalysis. DCT applied to JPEG compression has high resistance to attacks and good perceptual quality [1]. Hybrid approaches that integrate LSB and DCT trade capacity and robustness by varying based on local image features. Experimentations show enhanced payload capacity and security. Deep techniques such as encoder-decoder models surpass traditional by employing convolution structures with minimum distortion and maximum recovery loss functions. They have high PSNR and SSIM but preserve

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.

visual loyalty. The majority of them operate in both RGB and grayscale forms, adding greater flexibility. The use of adversarial networks as well as end-to-end training enhances embedding optimization and extraction performance as well as detection robustness [2].

Steganography is usually paired with encryption algorithms such as AES, ChaCha20, DES, and RSA. ChaCha20 and AES are more efficient and also quicker compared to traditional methods. Cryptography along with steganography guarantees confidentiality, integrity, and stealth. Spread Spectrum provides security but also a loss in image quality. Hybrid approaches frequently employ RSA for the safe exchange of AES keys. Pixel Value Differencing (PVD) using edge detection, e.g., the Canny algorithm, enhances security through concealment in less perceptible edges, maximizes embedding capacity at the cost of distortion [4]. Adaptive payload allocation provides performance improvement dependent on gradient intensity. Spatial domain difference expansion boosts payload capacity while preserving image quality. It employs pixel pair differences for reversible embedding, enhancing PSNR and MSE and accommodating domains such as legal and medical imaging [5]. Substitution of complete hidden images with LSB techniques enhances capability and fidelity with minimal computational burden [6].

These networks employ convolutional structures with loss functions tailored for minimum distortion and maximum recoverability. Deep steganography models provide better PSNR and SSIM, maintaining visual fidelity. Most of them support both grayscale and RGB, enhancing flexibility across domains. To provide safe transmission, steganography is coupled with encryption algorithms such as AES, ChaCha20, DES, and RSA [3].

Lastly, adding error correction to LSB-based techniques improves resistance under noisy transmission environments [7]. Combining LSB with cryptography and error correction provides data integrity and reliability, even in hostile environments, making the steganographic systems more resilient and usable.

The research in [8] examines LSB steganography with a focus on increasing extraction reliability via error correction techniques, as well as its combination with cryptography to increase data security. In [9], a deep learning model based on U-Net is proposed to enhance payload capacity and minimize visual hints, but it is still susceptible to adversarial attacks and steganalysis. An exhaustive survey in [10] categorizes deep learning steganography methods as cover-based and coverless, where coverless methods provide enhanced resistance against steganalysis upon training on databases such as COCO and ImageNet. The review in [11] gives a general overview of image steganography ranging from classical LSB to advanced neural network-based methods, highlighting issues concerning security, efficiency, and pixel-level data embedding. In [12], a hybrid method is presented, fusing cryptography with a modified LSB method on the blue channel of images, which maintains secure data transfer with little effect on computational complexity. An intuitive GUI for LSB steganography is implemented in [13], making embedding and decoding easy while adding AI for security and privacy improvement. Finally, [14] gives a comparative study of different steganographic methods, assessing their performance and providing advice on how to choose an appropriate method for a given application necessity.

Research works discuss the recent developments in audio and image steganography. A better spatial domain audio steganography method improves LSB stability with Root Mean Square (RMS) values, preserving data capacity in different audio formats [15]. A deep learning generative model, comprising a generator, extractor, and steganalyzer, trained with Mean Square Error (MSE), is up to 4% more accurate than traditional CNNs, particularly at low embedding rates [16]. In image steganography in spatial domain, chaotic maps such as Logistic and Chebyshev are employed to decide the embedding points, enhancing imperceptibility and modifying statistical features quantified using PSNR and MSE [17]. A new LSB technique employs randomness in pixel selection aided by the Collatz Conjecture modified and Diffie-Hellman Key Exchange, which boosts statistical attack resistance and enhances stego-image quality [18]. Another method combines AES-128 encryption with LSB, where image properties (e.g., resolution, file type) are used to decide RGB traversal order to enhance security without extra data embedding [19]. Finally, comparative study of image steganography algorithms like pseudo-random noise with Haar wavelets, Pixel Value Ordering (PVO), and Dual-Tree Complex Wavelet Transform (DT-CWT) reveals that increased use of higher bits enhances capacity but decreases PSNR, indicating that there is a trade-off between capacity and image quality [20].

One of the significant challenges in steganography research is achieving an optimal compromise among payload capacity, resilience, and perceptual quality. Adding capacity frequently comes at the expense of security or image/audio quality, while most traditional approaches, including simple LSB, are easily detectable using powerful steganalysis tools. Increasing security and resistivity to attack is a major challenge, as real-world implementations must also address computational overhead and resource constraints. Moreover, some methods have a hard time dealing with large quantities of input data efficiently because of system limitations, making their applicability in the real world even more challenging. Therefore, it is essential to conduct further research to overcome these issues and devise methods that attain a better balance among these fundamental properties.

II. Methodology

This section describes the entire methodology followed in this research to identify steganography in images through two different approaches: one involving frequency domain features through Discrete Cosine Transform (DCT) and the other through pixel-domain manipulations through Least Significant Bit (LSB) embedding. Both pipelines incorporate dimensionality reduction, statistical hypothesis testing, classification,

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.

ensemble learning, and an inference mechanism for real-world deployment.

The common methodology taken in this study is graphically outlined in Fig.1. The flowchart illustrates the overall pipeline for the DCT-based and LSB-based steganalysis, emphasizing fundamental steps like preprocessing of data, feature extraction (DCT parity analysis and pixel-level features), statistical testing, dimensionality reduction (SVD and Autoencoder), and training of models via varied classifiers and ensemble methods. The flowchart makes it easier to differentiate the almost similar strategies brought about by embedding nature in both steganographic techniques.

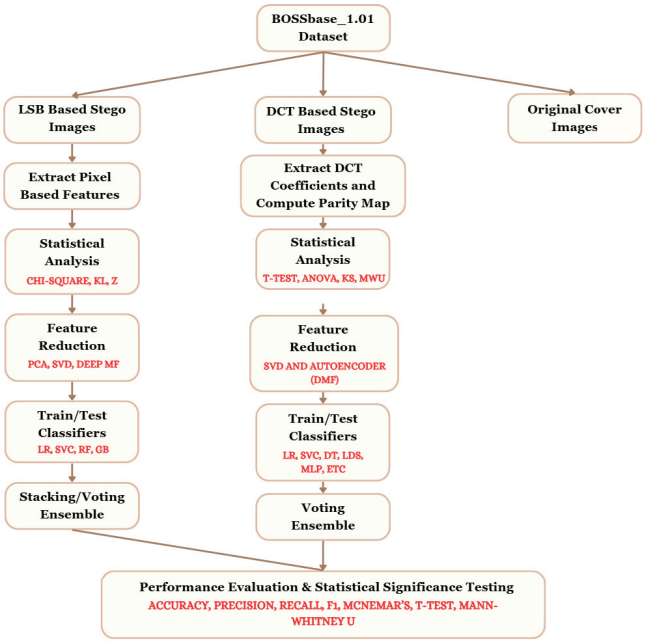

Fig. 1. Unified Methodology Flowchart for DCT and LSB Steganalysis

A. Dataset Overview

For both DCT-based and LSB-based experiments, the BOSSbase_1.01 dataset was utilized. This dataset comprises 10,000 grayscale images, each of size 256×256 pixels. The dataset was split into two main categories:

- Cover images: unmodified images which are originally in the dataset.
- Stego images: Images which were altered in order to embed hidden information using steganographic techniques (DCT or LSB embedding).

Each image is accompanied by a binary label indicating whether it is cover (0) or stego (1), allowing the problem to be treated as a supervised binary classification task.

B. Feature Extraction

1) DCT-Based Feature Extraction: The DCT method focuses on frequency-domain representation. Each image undergoes a 2D DCT transformation, and from the resulting DCT coefficients, parity features, which are sensitive to modifications caused by hidden data are extracted. These features capture subtle changes in frequency distributions that are often imperceptible to the human eye.

Two techniques were used for dimensionality reduction:

- Singular Value Decomposition (SVD): Compresses the parity feature vectors while preserving the most informative patterns.
- Autoencoder: A deep learning-based encoder-decoder model which is trained to learn compressed representations. Only the encoder's latent outputs were used for classification.

2) LSB-Based Feature Extraction: For the LSB pipeline, images were processed in the spatial (pixel) domain. Each 256×256 image was first flattened into a 1D vector. These high-dimensional vectors were then reduced using:

- Principal Component Analysis (PCA): Retains only the principal directions of variance.
- SVD: Similar to the DCT pipeline, used to reduce data size and complexity.
- Deep Matrix Factorization (DMF): Implemented using a pre-trained autoencoder model (`encoder.h5`) to generate compact, non-linear feature vectors.

All feature vectors were saved in `.npy` format for efficient reuse.

C. Statistical Testing on DCT Features

To validate the extracted DCT features, the following statistical tests were applied to evaluate differences between cover and stego classes:

- T-Test: Tested for differences in mean values.
- Mann-Whitney U Test: Non-parametric test for distributional differences.
- ANOVA: Assessed variation across multiple groups.
- Kolmogorov-Smirnov (KS) Test: Measured empirical distribution differences.

Results showed that over 90% of features had statistically significant differences, confirming their relevance for classification.

D. Classifier Training and Evaluation

Both pipelines trained machine learning classifiers on the reduced feature vectors. For DCT, classifiers were trained on both SVD and Autoencoder-reduced features. For LSB, classifiers used PCA, SVD, and DMF features.

- Classifiers Used: A wide range of traditional and ensemble machine learning classifiers were employed, including Support Vector Machine (SVM), Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, K-Nearest Neighbors (KNN), Gaussian and Bernoulli Naive Bayes, Linear Discriminant Analysis (LDA), and Multi-layer Perceptron (MLP).
- Evaluation Metrics: Model performance was assessed using multiple evaluation metrics such as Accuracy, Precision, Recall, ROC AUC, and the Confusion Matrix

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.

to provide a comprehensive analysis of classification effectiveness.

All trained models were saved using `joblib` for future inference.

E. Deep Learning Classifier for LSB

A Convolutional Neural Network (CNN) was developed using Keras for the LSB pipeline. The model included:

- Multiple dense layers with ReLU activation
- Dropout and Batch Normalization
- Softmax output for binary classification

The model used features from PCA, SVD, and DMF. Early stopping was used to mitigate overfitting. CNN performance was limited by a lack of spatial information.

F. Ensemble Learning

1) Voting Classifier (DCT): Soft voting was employed to combine predictions from top classifiers using SVD and Autoencoder features.

2) Stacking Classifier (LSB): The stacking model included:

- Base learners: GaussianNB, BernoulliNB, Logistic Regression, XGBoost
- Meta learner: Logistic Regression

This ensemble achieved the highest accuracy in the LSB pipeline.

G. Statistical Analysis (LSB Pipeline)

Unsupervised tests were also performed:

- Chi-square Test: Compared pixel intensity histograms.
- KL Divergence: Measured divergence from reference image distributions.
- Z-Test: Compared pixel mean differences.

These methods provided auxiliary confidence for the classification results.

H. Inference Pipeline and Deployment

A real-time inference system was implemented:

1) Upload: User submits PNG or JPG image.
2) Preprocessing: Image is converted to PGM, resized to 256×256, and normalized.
3) Feature Extraction: DMF features extracted using autoencoder.
4) Classification: SVM model classifies image.
5) Statistical Tests (optional): Chi-square, KL, Z-Test results shown.
6) Output: Final prediction with probability.

III. Results and Discussion

A. DCT-Based Steganalysis

1) Statistical Significance Testing: The parity features derived from Discrete Cosine Transform (DCT) were subjected to multiple hypothesis testing techniques to assess their sensitivity to steganographic embedding. The outcomes are summarized below:

- T-Test: 919 out of 1000 features showed significant difference ($p < 0.05$).
- Mann-Whitney U Test: 919 out of 1000 features with $p < 0.05$.
- ANOVA: 919 out of 1000 features with $p < 0.05$.
- Kolmogorov-Smirnov Test: 895 out of 1000 features with $p < 0.05$.

The results being consistent across all four tests indicate that over 90% of the parity features are statistically different between stego and cover images. This confirms that the embedding method induces detectable changes in the frequency domain, without requiring access to the original message or decoder.

2) Classification with SVD Features: After dimensionality reduction using Singular Value Decomposition (SVD), various machine learning classifiers were evaluated. The performance metrics are reported in Table I.

TABLE I
Classification Performance using SVD Features for DCT Embeddings

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	0.6987	0.6828	0.7314	0.7810
SVC	0.6210	0.6087	0.6572	0.7008
Decision Tree	0.5002	0.4955	0.5033	0.5003
Random Forest	0.4605	0.4545	0.4462	0.4757
Gradient Boosting	0.6522	0.6455	0.6608	0.7328
AdaBoost	0.6498	0.6436	0.6562	0.7155
KNN	0.4218	0.4179	0.4266	0.4071
Naive Bayes	0.6825	0.6703	0.7062	0.7524
LDA	0.6997	0.6811	0.7405	0.7794
MLPClassifier	0.6182	0.6173	0.6032	0.7276
XGBoost	0.5890	0.5828	0.5987	0.6656

Linear models (Logistic Regression, LDA) outperform complex tree-based models, indicating that SVD provides a linearly separable subspace. Ensemble models and deep networks are underperforming, likely due to the loss of spatial information in the transformed features.

3) Classification with Autoencoder Features: An autoencoder was trained to compress and reconstruct the DCT features. The training loss stabilized after 50 epochs, as shown in Fig. 2

The extracted latent features were classified using the same classifiers and are the results are illustrated in Table II.

TABLE II
Classification Performance using Autoencoder Features

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	0.6737	0.6576	0.7118	0.7460
SVC	0.6342	0.6075	0.7390	0.7083
Decision Tree	0.5282	0.5237	0.5240	0.5282
Random Forest	0.4805	0.4754	0.4735	0.5304
Gradient Boosting	0.6465	0.6259	0.7113	0.7264
AdaBoost	0.6278	0.6247	0.6219	0.6941
KNN	0.4310	0.4289	0.4493	0.4355
Naive Bayes	0.5610	0.5447	0.6926	0.6289
LDA	0.6755	0.6582	0.7173	0.7452
MLPClassifier	0.6158	0.6081	0.6305	0.7210
XGBoost	0.5900	0.5809	0.6179	0.6682

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.

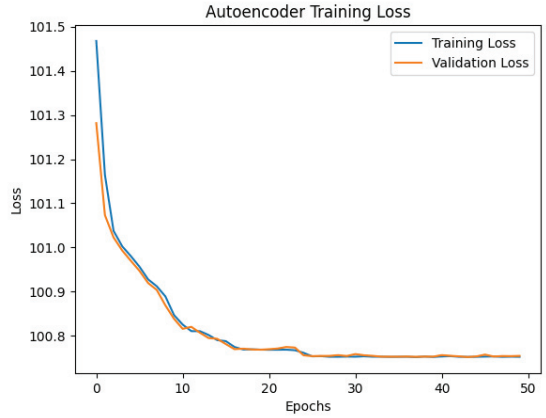

Fig. 2. Autoencoder Training and Validation Loss Curve

The performance was similar to SVD-based features, with Logistic Regression and LDA again being the best. The voting classifier yielded the following.

- SVD Features: Accuracy = 0.619, AUC = 0.6899
- Autoencoder Features: Accuracy = 0.6023, AUC = 0.6781

Hence, while both dimensionality reduction techniques are usable, SVD marginally outperforms the autoencoder in classification accuracy and AUC. This is primarily due to SVD's ability to capture global structures in the data with fewer parameters and less computational complexity. In contrast, the autoencoder, despite its flexibility in learning complex nonlinear relationships, may struggle with overfitting when dealing with small datasets or insufficient regularization. Consequently, SVD provides a more stable and interpretable solution for dimensionality reduction in this context.

B. LSB-Based Steganalysis

1) Traditional and Generative Models: Table III presents the performance of various classifiers in the detection of LSB steganography. Traditional models like Naive Bayes and Logistic Regression achieved modest accuracies in the range of 44–46%. Probabilistic generative models, including Gaussian Mixture Models (GMM) and Hidden Markov Models (HMM), showed slight improvements with accuracies approaching 50%, suggesting the presence of underlying probabilistic structures that are only weakly captured.

2) Advanced Models and Ensembles: Boosting-based models such as XGBoost, CatBoost, and LightGBM failed to generalize, all performing near chance level (10–11%). In contrast, AdaBoost achieved up to 49.6% due to its simplicity and resistance to overfitting. The standout performer was the stacking ensemble model, achieving 89.89% accuracy, leveraging multiple classifiers to capture diverse data patterns. This demonstrates that combining weaker models significantly enhances robustness and generalization.

TABLE III
LSB-BASED STEGANALYSIS ACCURACY

Model	Accuracy (%)
Gaussian Naive Bayes	43.88
Bernoulli Naive Bayes	45.88
Logistic Regression	44.93
Gaussian Mixture Model	50.10
Hidden Markov Model	50.45
Convolutional Neural Network	19.86 – 20.53
XGBoost	10.16
XGBoost (Tuned)	41.58
CatBoost	10.09 – 10.11
AdaBoost	44.86 – 49.60
LightGBM	10.66
Stacking Ensemble	89.86 – 89.89

3) Statistical Testing:

- Chi-square Test: $p = 1.0000$, indicating no global histogram shift.
- KL Divergence: 1.0786, reflecting moderate divergence in distributions.
- Z-test: $p = 0.0000$, confirming localized pixel-level alterations.

These results suggest LSB embedding introduces subtle but statistically significant changes detectable through localized analysis.

4) Final Prediction Insight: Fig.3. depicts the UI for the user to upload an image and the predict the results to check if it is a stego image or a cover image.

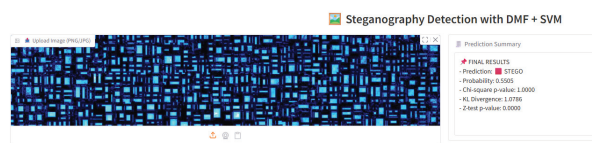

Fig. 3. A Detection Framework to Classify Uploaded Images as Stego or Cover

The ensemble classified the image as stego with a probability of 0.5505, a moderate confidence. This aligns with statistical findings and validates the model's predictive capability. Besides the probabilistic prediction of the model, histogram analysis was performed to plot the grayscale pixel intensity distribution of the input image, as presented in Fig. 4.

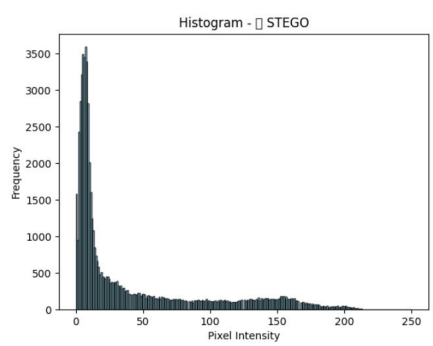

Fig. 4. Statistical deviations introduced by steganographic embedding.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.

The analysis confirms the hypothesis that steganographic embedding causes minor statistical anomalies in the image. The histogram was tested against a uniform reference distribution using the Chi-square goodness-of-fit test, the Kullback-Leibler divergence was calculated to measure distributional divergence, and a Z-test was conducted in comparison to average cover image statistics (mean approximately 127, std approximately 40). These statistical measures, in conjunction with the confidence score of the classifier, provide evidence at large for stego images to be differentiated from clean covers.

IV. Conclusion

This study presents a comprehensive steganalysis framework targeting both DCT-based and LSB-based steganographic embedding techniques. For the DCT-based approach, we extracted parity features from the frequency domain and verified their statistical separability through multiple hypothesis tests, including T-Test, Mann-Whitney U, ANOVA, and Kolmogorov-Smirnov. Dimensionality reduction via SVD and Autoencoders yielded compact representations that, when classified using various machine learning models, demonstrated that linear classifiers such as LDA and Logistic Regression consistently outperformed more complex models. This suggests that the transformed features possess a linearly separable structure conducive to effective classification.

In the LSB-based steganalysis, although traditional statistical models and boosting classifiers showed limited performance, ensemble methods—particularly stacking—proved significantly more effective, achieving a classification accuracy of nearly 90%. This highlights the value of model aggregation in scenarios with weak or noisy features. Statistical tests such as Chi-square, KL Divergence, and Z-tests further substantiated the detectability of steganographic alterations.

Overall, the dual-path steganalysis methodology demonstrates that both spatial and frequency domain features, when appropriately processed and combined with rigorous statistical testing and robust classification pipelines, can yield reliable detection of hidden information. Future work may explore hybrid feature sets and deep learning-based fusion strategies to further enhance detection robustness and generalizability across varied steganographic algorithms.

Acknowledgment

The authors wish to thank Amrita Vishwa Vidyapeetham from the bottom of their hearts for extending the required support, resources, and research infrastructure that helped them complete this research work successfully. We also thank the encouragement and support provided while preparing this publication.

References

[1] B. A. Dharani, B. Yashaswini, G. R. S. Reddy, and S. M. Rajagopal, "Multimodal Steganography: A Comparative Analysis of LSB and DCT Methods for Image and Audio Data Concealment," *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, pp. 1–5, 2024.

[2] R. Varman and P. Malathi, "Image Data Hiding Scheme Using Adversarial Embedding," *2023 Innovations in Power and Advanced Computing Technologies (i-PACT)*, pp. 1–7, 2023.

[3] M. Kataria, K. Jain, and N. Subramanian, "Exploring advanced encryption and steganography techniques for image security," *2023 11th International Symposium on Digital Forensics and Security (ISDFS)*, pp. 1–6, 2023.

[4] B. Vishnu, L. V. Namboothiri, and S. R. Sajeesh, "Enhanced image steganography with PVD and edge detection," *2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 827–832, 2020.

[5] I. F. Ramadhan, R. D. A. Anandha, A. W. C. D'Layla, N. J. De La Croix, and T. Ahmad, "Image Steganography using Customized Differences between the Neighboring Pixels," *2024 7th International Conference on Informatics and Computational Sciences (ICICoS)*, pp. 496–501, 2024.

[6] G. S. Kishore, Y. L. Kumar, K. S. Abhilash, and E. Aparna, "A Bitwise Operations-Based Image Steganography Technique for LSB Replacement," *2024 1st International Conference on Sustainable Computing and Integrated Communication in Changing Landscape of AI (ICSCAI)*, pp. 1–4, 2024.

[7] P. Kanjanamek, N. Chaiyabud, N. Kitikhungumjon, and S. Fugkeaw, "An Adaptive Cloud-Based Image Steganography System with Fast Stego Retrieval," *2024 16th International Conference on Knowledge and Smart Technology (KST)*, pp. 29–34, 2024.

[8] R. Srusti and M. L. J. Shruthi, "Implementation and Comparative Analysis of CNN and Discrete Haar Wavelet Transform in Image Steganography," *2024 IEEE 16th International Conference on Computational Intelligence and Communication Networks (CICN)*, pp. 912–916, 2024.

[9] M. I. Hossain, S. Kadir, F. I. Fagun, I. Samiul, R. Z. Saukhin, and M. I. Hossain, "Enhanced CNN approaches for multi-image embedding in image steganography," *2024 2nd International Conference on Information and Communication Technology (ICICT)*, pp. 100–104, 2024.

[10] Y. Wencui, T. Zhu, and Y. Liu, "A Review of Deep Learning Based Image Steganography Methods," *2024 6th International Conference on Natural Language Processing (ICNLP)*, pp. 630–637, 2024.

[11] S. Hegde, P. Sunag, R. P. Varun, et al., "Exploring the Effectiveness of Steganography Techniques: A Comparative Analysis," *2023 3rd International Conference on Smart Data Intelligence (ICSMDI)*, pp. 181–186, 2023.

[12] I. Shukla, A. Joshi, and S. Girme, "LSB Steganography Mechanism to Hide Texts Within Images Backed with Layers of Encryption," *2023 16th International Conference on Security of Information and Networks (SIN)*, pp. 1–6, 2023.

[13] M. Meenadshi, et al., "AI-Enhanced LSB Steganography Interface: Concealed Data Embedding Framework," *2023 9th International Conference on Smart Structures and Systems (ICSSS)*, pp. 1–4, 2023.

[14] N. Saxena and H. Kumar, "Steganography Techniques a Review," *2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)*, pp. 1355–1360, 2023.

[15] A. G. Devi, A. Thota, G. Nithya, S. Majji, A. Gopatoti, and L. Dhavamani, "Advancement of digital image steganography using deep convolutional neural networks," *2022 International Interdisciplinary Humanitarian Conference for Sustainability (IIHC)*, pp. 250–254, 2022.

[16] J. Qiu, "Generative image steganography scheme based on deep learning," *2022 International Conference on Education, Network and Information Technology (ICENIT)*, pp. 191–194, 2022.

[17] M. C. Kasapbaşı and İ. Bayam, "A new improved LSB chaotic image steganography scheme," *2022 Conference of Russian Young Researchers in Electrical and Electronic Engineering (ElConRus)*, pp. 332–335, 2022.

[18] A. D. Molato, F. B. Calanda, A. M. Sison, and R. P. Medina, "LSB-based random embedding image steganography technique using modified Collatz conjecture," *2022 7th International Conference on Signal and Image Processing (ICSIP)*, pp. 367–371, 2022.

[19] R. Dumre and A. Dave, "Exploring LSB steganography possibilities in RGB images," *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–7, 2021.

[20] J. Singh and M. Singla, "A novel method of high-capacity steganography technique in double precision images," *2021 International Conference on Computational Performance Evaluation (ComPE)*, pp. 780–784, 2021.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:56 UTC from IEEE Xplore. Restrictions apply.